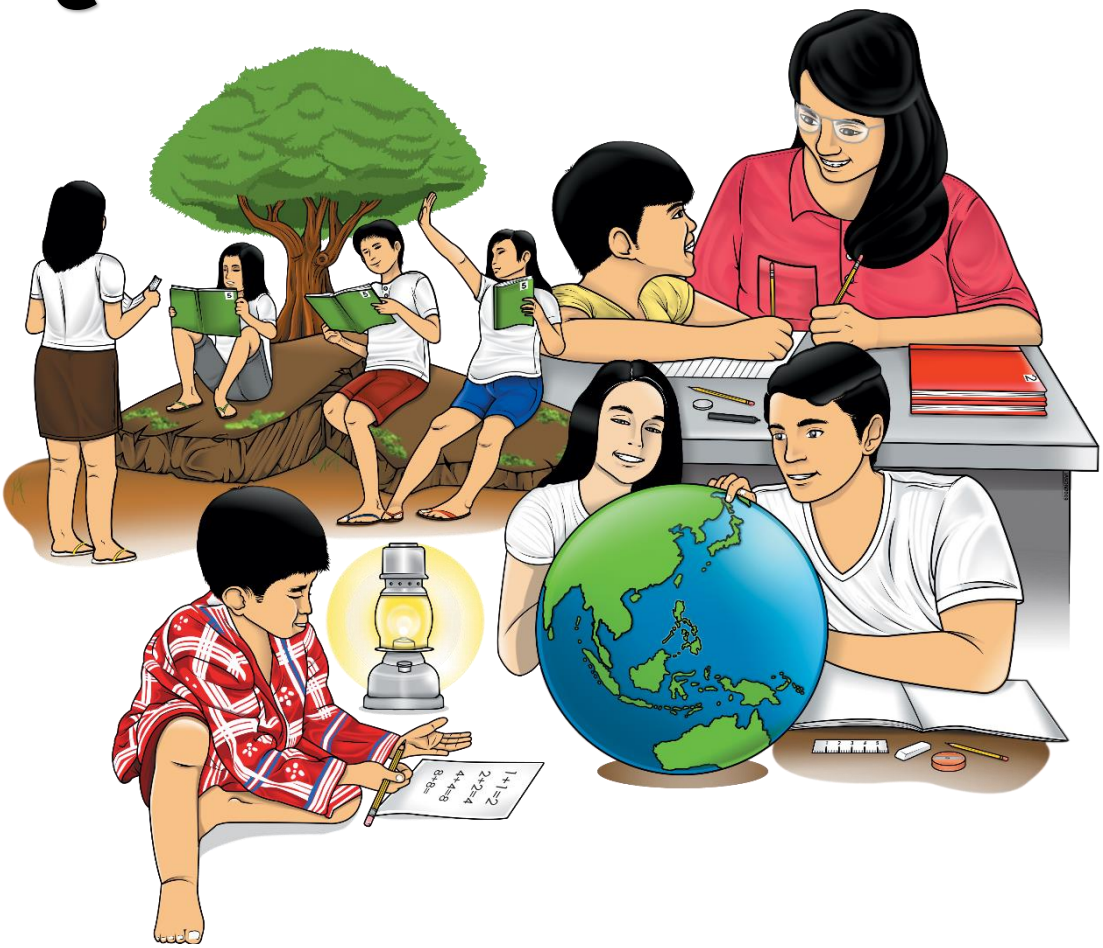
Senior High School

GENERAL BIOLOGY I

Quarter 2 – Module 1



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What I Need to Know

Photosynthesis is the process by which plants use sunlight, water, and carbon dioxide to manufacture oxygen and energy in the form of sugar. It absorbs by the body through ingestion which becomes usable materials into cellular energy and respiration process.

At the end of the lesson the learners are expected to:
Differentiate aerobic from anaerobic respiration (STEM-BIO 12-IIa-j-6)

What I Know

Directions: Choose the letter of the correct answer.

_____1. Which of the following sentences describe the process of photosynthesis?
A. Plants absorb water through their roots.
B. Plants make chlorophyll, the pigment in the leaves.
C. Plants take in carbon dioxide and release oxygen.
D. Plants use light energy to transform water and carbon dioxide into glucose and oxygen.

_____2. What are the products of photosynthesis?
A. oxygen and water
B. glucose and oxygen
C. glucose and chlorophyll
D. water and carbon dioxide

_____3. The chemical formula for glucose is $C_6H_{12}O_6$. What elements do you think make up glucose?
A. carbon and oxygen
B. hydrogen and oxygen
C. carbon and hydrogen
D. carbon, hydrogen, and oxygen

_____4. Which is TRUE about the sugar molecules in plants?
A. Molecules come from the soil.
B. One of the many food sources of plants.
C. Molecules are the result of a chemical reaction.
D. It is made from molecules of water and minerals.

_____5. Which raw materials are needed by plants to make food?
A. air and soil
B. soil and water
C. minerals and water
D. carbon dioxide and water

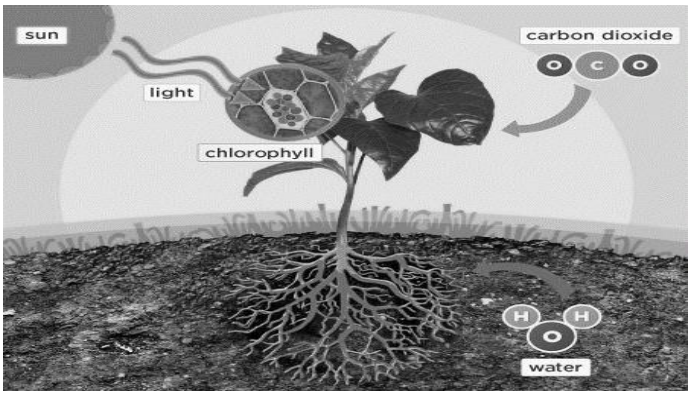
Lesson 1	*Photosynthesis and Respiration*

What's New

Activity:

Photo-Pic

Directions: Study the diagram and answer the guided questions. Write your answer in a separate sheet of paper.



Guided Questions:

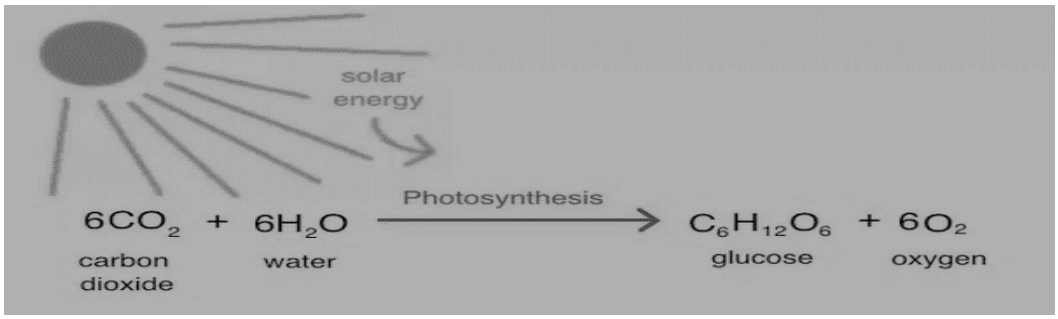
1. How does the plant absorb water from the soil?
2. How does plant absorb carbon dioxide from the air?
3. What absorbs the light energy from the sun?
4. Explain how photosynthesis works.

What Is It

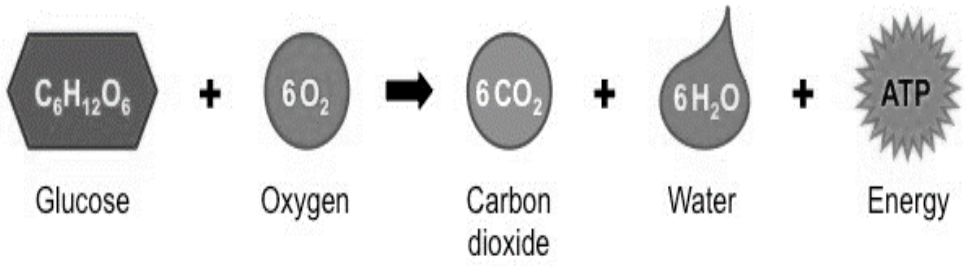
Energy conversions involve the release of energy stored in chemical bonds. The energy storage molecules used by all cells is called **Adenosine Triphosphate (ATP)**. It is composed of adenine, ribose and three phosphates; stores energy, while some molecules exchange energy.

ATP consists of three major parts:

- **Ribose** is a five-carbon sugar molecule
- **Adenine** is one of the nucleotides of **RNA** and **DNA**. The **ribose bonds with adenine** to form a molecule called **adenosine**.
- The **three phosphates**, called **triphosphate group** are linked in a chain that is bonded to adenosine. When these bonds are broken, the energy stored in the bonds is transferred to other chemical reactions.



During photosynthesis, energy is exchanged when chemical reactions called **reduction reactions** add electrons to carbon dioxide molecule. **Reduction** is the addition of electrons to an atom or ion. It is called a reduction process because of the addition of electrons, which carry a negative charge, reduces the charge of the molecule. The addition of an electron increased the energy contained in the molecule. The reduction of carbon dioxide molecules leads to the formation of energy rich carbohydrates, including glucose. Glucose is the molecule used as an energy source by almost all organisms. During photosynthesis and respiration, certain molecules act as electron carriers, transferring energy in the form of high- energy electrons that are repeatedly gained and lost. The lost of an electron from an atom or ion is called **oxidation.** Oxidized molecules are low in energy. Although oxygen is an electron



acceptor, many oxidation reactions do not involve oxygen. Chemical reactions in which one molecule is oxidized and the other is reduced are called **oxidation-reduction reactions**, or simply called **redox reactions**. There are three important events that occur during the process of photosynthesis, these are light energy absorption, conversion into chemical energy and formation of carbohydrates by reduction of carbon dioxide.

Cell respiration is a process by which food molecules are broken down to release energy. In any cellular processes involve, energy is required to both build and breakdown molecules like large proteins that make up the muscles are actively built from smaller molecules. Complex carbohydrates are broken down into simple sugars that the cell uses for energy and exporting toxic wastes.

What's More

Activity

Which Is Which?

Directions: Complete the table.

	Photosynthesis	**Respiration**
Raw Materials		
Cell Organelle		
Energy Involved		
End Products		

What I Can Do

ATP- Food for Thought:

Directions: Read the paragraph and answer the question.

ATP- Food for Thought: Different foods contain different amount of stored energy. The amount of energy in food is determined by the relative quantities of protein, fat, carbohydrate, and water in the food. Fruits and vegetables contain large amount of water and low calories. When a human consumes more energy than it needs, the excess is converted into fat for long-term storage. If you are into diet, why is it advisable to eat more fruits and vegetables and do exercise?

Assessment

Directions: Identification.

_____1. It the conversion of light energy to food energy.
_____2. It is an addition of electrons to an atom or ion.
_____3. It is the loss of an electron from an atom or ion.
_____4. It is energy storage molecule used by most organisms.
_____5. It is the process by which food molecules are broken down to release energy.

What I Need to Know

Plants need sunlight to energize the green pigment, chlorophyll found in the leaf for them to undergo photosynthesis with carbon dioxide and water as other essential component for the process to continue.

At the end of the lesson the learners are expected to:

1. **Explain the functions of chlorophyll and other photosynthetic pigments (STEM_BIO11/12-IIa-j-3)**; and
2. **Identify the events that took place during light reaction**. **(STEM_BIO11/12)-IIa-j-4**

What I Know

Directions: Write the letter of the correct answer.

_____1. Which of the following is NOT a product of light reactions?
A. ATP B. oxygen C. NADPH D. glucose(sugar)

_____2. What are the two products of the light reactions needed for the Calvin cycle?
A. NAD and NADPH C. O_2 and NADPH
B. NADPH and ATP D. NADP+ and glucose

_____3. Which organelle performs photosynthesis?
A. nucleus B. chloroplast C. lysosomes D. mitochondria

_____4. What are the reactants of light reactions phase?
A. water and CO_2 C. sunlight and water
B. oxygen and water D. sunlight and oxygen

_____5. What is the purpose of the light dependent reactions?
A. to make sugars

B. to reflect green light
C. to capture energy and make sugar
D. to provide the energy for the Calvin cycle

Lesson 2 — *Chlorophyll and other Photosynthetic Pigments*

What's New

Activity A:

HAI-CHLO

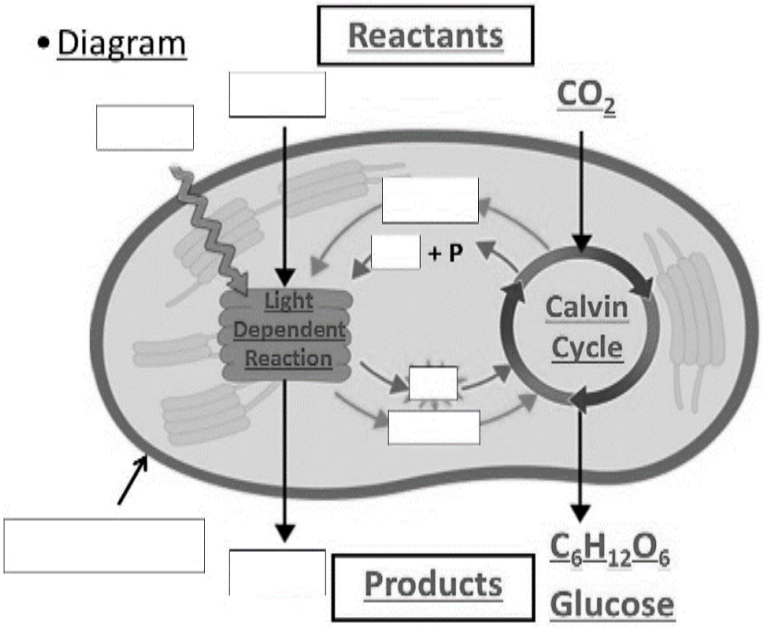
1. Write a 3-line poem (Haiku) 5-7-5 syllables about the role of chlorophyll.
2. Make sure it is authentic meaning your own. Afterwards, upload it to your social media account or write in a separate paper.
3. Follow the haiku rubrics found at the last page of the module.

Activity B:

FLIP-LABEL ME!!!

Directions: Cut the words inside the box and paste them in the diagram.

LIGHT	H_2O
NADP	ATP
O_2	NADPH
ADP	CHLOROPLAST



What Is It

Types of Photosynthetic Pigments:

•**Chlorophyll a**- appears green in color. It absorbs blue and red light and reflects green light. It is the most abundant type of pigment in leaves and thus the most important type of pigment in chloroplast. At a molecular level it has a porphyrin ring that absorbs light energy.

•**Chlorophyll b** – less abundant than chlorophyll a but has ability to absorb a wider wavelength of light energy.

•**Chlorophyll c** – is not found in plants but is found in some microorganisms capable of performing photosynthesis.

•**Carotenoid and Phycobillin** – Carotenoids are found in many photosynthetic organisms as well as in plants. They absorb light between 460 to 550 nm and hence appear orange, red and yellow. Phycobillin, a water-soluble pigment, is found in chloroplast. (Flint, 2019)

Light dependent reaction used photosynthetic pigments (organized into photosystems) to convert light energy into chemical energy (specifically ATP and NADPH). It occurs in the grana (stacks of thylakoids) within the chloroplast and involve three steps

STEP 1: Excitation of photosystems by light energy

- Photosystems are groups of photosynthetic pigments (including chlorophyll) embedded within the thylakoid membrane
- Photosystems are classified according to their maximal absorption wavelengths (PS I= 700 nm; PS II = 680 nm)
- When a photosystem absorbs light energy, delocalized electrons within the pigments become energized or 'excited'
- These excited electrons are transferred to carrier molecules within the thylakoid membrane.

STEP 2: Production of ATP via an Electron Transport Chain

- Excited electrons from Photosystem II (P680) are transferred to an electron transport chain within the thylakoid membrane
- As the electrons are passed through the chain, they lose energy, which is used to translocate H+ ions into the thylakoid
- This build-up of protons within the thylakoid creates an electrochemical gradient, or proton motive force
- The H+ ions return to the stroma (along the proton gradient) via the transmembrane enzyme ATP synthase (chemiosmosis)
- ATP synthase uses the passage of H+ ions to catalyze the synthesis of ATP (from ADP + Pi)
- This process is called photophosphorylation – as light provided the initial energy source for ATP production
- The newly de-energized electrons from Photosystem II are taken up by Photosystem I

STEP 3: Reduction of NADP+ and Photolysis of Water

- Excited electrons from Photosystem I may be transferred to a carrier molecule and used to reduce NADP+
- This forms NADPH – which is needed (in conjunction with ATP) for the light independent reactions
- The electrons lost from Photosystem I are replaced by de-energized electrons from Photosystem II
- The electrons lost from Photosystem II are replaced by electrons released from water via photolysis
- Water is split by light energy into H+ ions (used in chemiosmosis) and oxygen (released as a by-product)

What's More

Activity

CHLOROPHYLL PAINTING

Materials

Small metal spoon or stone, 10 or more available green leaves, white sheet of paper

Preparation

1. Fold a sheet of white paper into halves.
2. Insert several leaves between the folded paper.
3. Press firmly the folded paper with leaves.
4. Rub the surface with the metal spoon.

Guided Questions:

1. Is there any mark on the paper? _____
2. Describe what you observed. _____
3. What does the marks demonstrate? _____

What I Can Do

Directions: Explain briefly.

How do chlorophyll and other pigments help the plants to survive?

_____

Assessment

Directions: Write the letter of the correct answer.

_____1. Which organelle performs photosynthesis?

A. Nucleus B. Chloroplast C. Lysosomes D. Mitochondria

_____2. Which of the following is NOT a product of light reactions?

A. ATP B. oxygen C. NADPH D. glucose (sugar)

_____3. What are the two products of the light reactions needed for the calvin cycle?

A. NAD and NADPH C. O_2 and NADPH

B. NADPH and ATP D. NADP+ and glucose

_____4. What is the purpose of the light dependent reactions?

A. to make sugars

B. to reflect green light

C. to capture energy and make sugar

D. to provide the energy for the Calvin cycle

_____5. What are the reactants of light reactions phase?

A. water and CO_2 C. sunlight and water

B. oxygen and water D. sunlight and oxygen

What I Need to Know

Calvin Cycle which is the second stage of photosynthesis can proceed without light, so its steps are sometimes called "light-independent" or "dark" reactions. It builds carbohydrates from smaller molecules and is an energy-consuming process. ATP is used as an energy source and NADPH as a reducing agent to add high-energy electrons to synthesize the sugar.

At the end of the lesson the learners are expected to:

Describe the significant events of the Calvin cycle. (STEM-BIO12-IIa-j-5)

What I Know

Directions: Write A if you agree with the statement and D if you disagree.

_____1. The Calvin cycle can happen at night.

_____2. Oxygen is NOT a reactant of the Calvin cycle.

_____3. Glucose is one of the products of Calvin cycle.

_____4. Calvin cycle occurs in the mitochondria.

Lesson 3 — *Phases of Calvin Cycle*

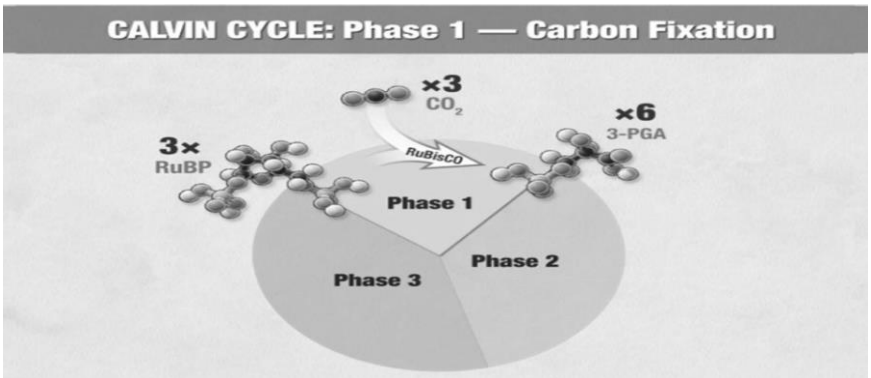
What's New

Activity 1

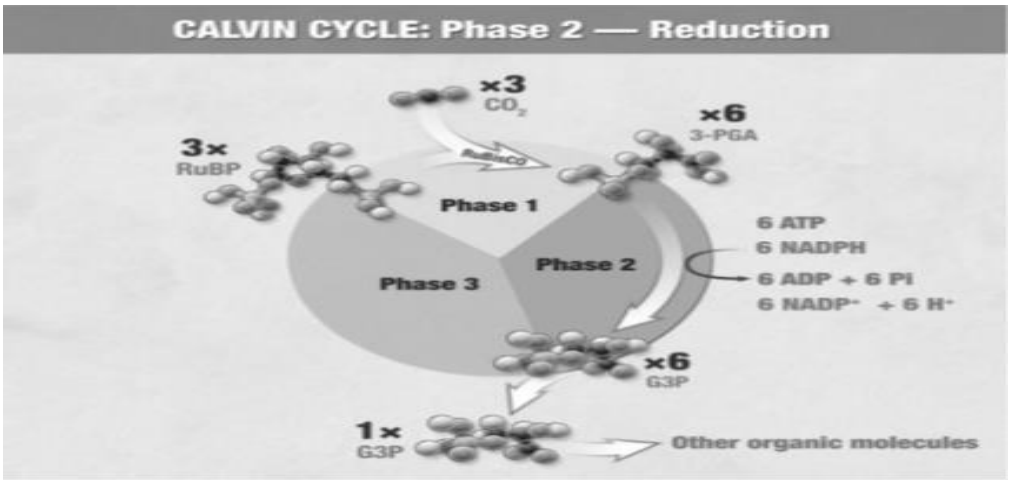
Phases of Calvin Cycle

Directions: Supply the needed information in each diagram.

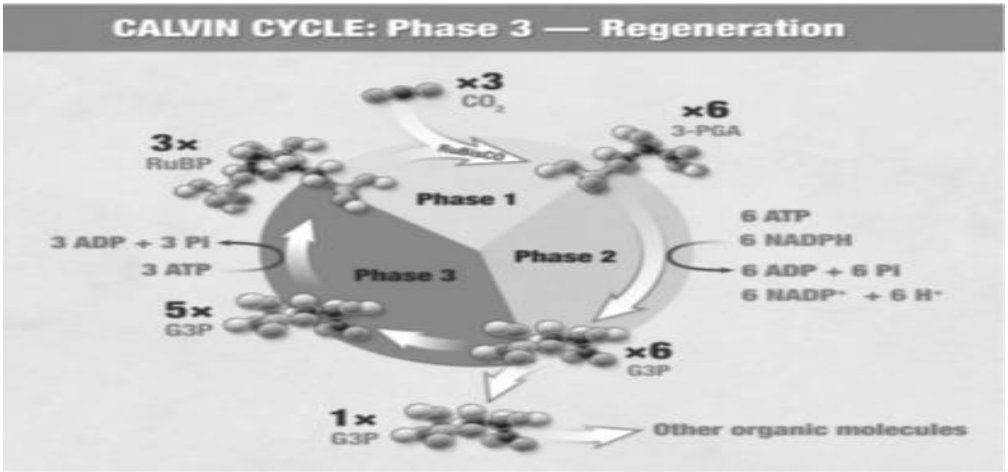
Directions: Label the diagram with the three phases of the Calvin cycle.



1. Describe the diagram. _____
2. What enzyme catalyzes the reaction in this phase? _____



Describe the diagram. _____



1.Describe the diagram. _____
_____

2.Why are the series of reactions in the Calvin cycle called a "cycle"?
_____

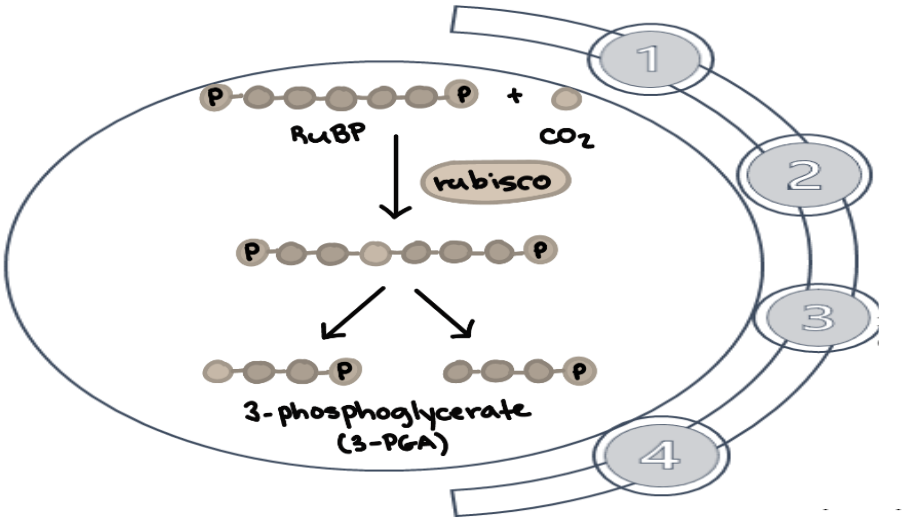
3.At the end of the Calvin cycle, what molecules have the energy that originally came from light? _____

What is It

The **light-independent reaction** of photosynthesis is a chemical reaction that converts carbon dioxide and other compounds into glucose. This reaction occurs in the stroma, the fluid-filled area of a chloroplast outside of the thylakoid membranes. It takes the light-dependent reaction and performs further chemical processes on them.

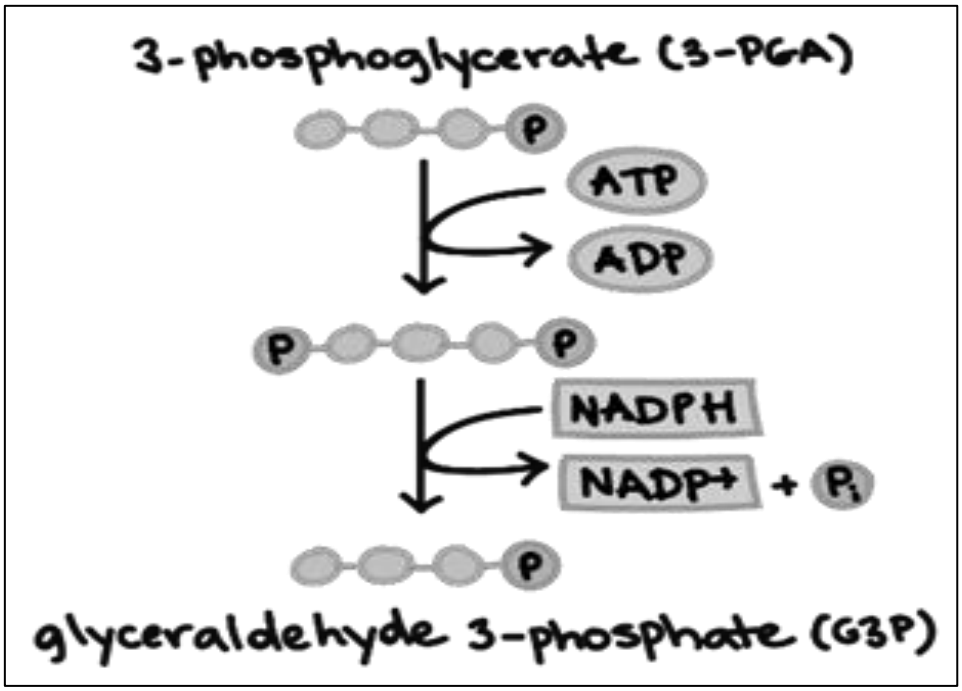
For the Calvin cycle to synthesize one molecule of sugar (G3P), three molecules of CO_2 must enter the cycle. The cycle is divided into three phases:

PHASE 1 – CARBON FIXATION



- The Calvin cycle begins when each molecule of CO_2 is attached to a five-carbon sugar, ribulose biphosphate (RuBP).
- This reaction is catalyzed by the enzyme RuBP carboxylase (rubisco)- the most abundant protein on Earth.
- The product of this reaction is an unstable six-carbon intermediate that immediately splits into two molecules of 3-phosphogylcerate.
- For every three CO_2 molecules that enter the Calvin cycle via rubisco, three RuBP molecules are carboxylated forming six molecules of 3-phosphoglycerate.

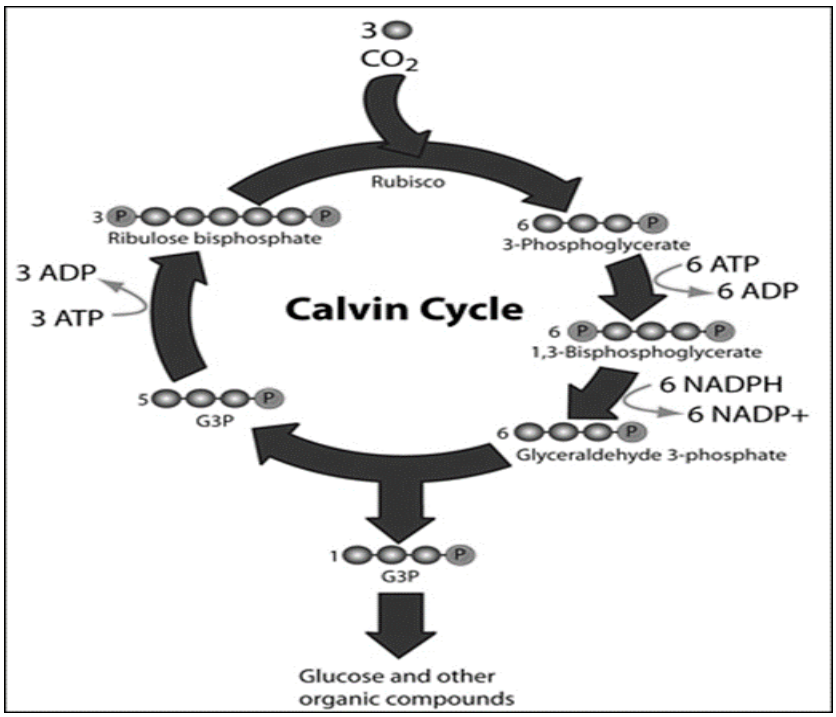
PHASE 2– REDUCTION



This endergonic reduction phase is a two-step process that couples ATP hydrolysis with the reduction of 3-phosphoglycerate to glyceraldehyde phosphate.

1. An enzyme phosphorylates 3-phosphoglycerate by transferring a phosphate group from ATP. This reaction,
    - Produces 1, 3-biphosphoglycerate
    - Uses six ATP molecules of 1,3-biphosphoglycerate.
    - Primes 1,3-biphosphoglycerate for the addition of high energy electrons from NADPH
2. Electrons from NADPH reduce the carboxyl group of 1,3-biphosphoglycerate to aldehyde group of glyceraldehyde-3-phosphate(G3P)
    - The product, G3P, stores more potential energy than the initial reactant, 3-phosphoglycerate.
    - G3P is the same three carbon sugar produces when glycolysis splits glucose.
3. For every three CO_2 molecules that enter the Calvin Cycle, six G3P molecules are produces only one of which can be counted as net gain.
    - The cycle begins with three five-carbon RuBP molecules-a total of 15 carbons
    - The six G3P molecules produced contain 18 carbons, a net gain of three carbons from CO_2.
    - One G3P molecule exits the cycle, the other five are recycled to regenerate three molecules of RuBP.

PHASE 3– REGENERATION OF STARTING MATERIAL (RuBP)



1. A complex series of reactions rearranges the carbon skeletons of five G3P molecules into three RuBP molecules.
    - These reactions require three ATP molecules

- RuBP is thus regenerated to begin the cycle again.

2. For the net synthesis of one G3P molecule, the Calvin cycle uses the products of the light reactions:
   - 9 ATP molecules and 6 NADPH molecules
3. G3P produced by the Calvin cycle is the raw material used to synthesize glucose and other carbohydrates.
4. The Calvin cycle uses 18 ATP and 12 NADPH molecules to produce one glucose molecule.

Every living thing on Earth depends on the Calvin cycle. Plants and other organisms including herbivores depend on the Calvin cycle for energy and food. Even organisms that eat other organisms, such as carnivores, depend on the Calvin cycle. Without it, they wouldn't have the food, energy, and nutrients they need to survive.

What's More

Directions: Pick the correct term from the word pool to complete the paragraph. Words can be used more than once.

Three molecules of 1. _____ combine with 2. _____ with the help of the 3. _____enzyme. This forms an unstable intermediary compound that splits to form six molecules of 4. _____. These molecules combine 5. _____ with group from 6. _____ to form 7_____. 8. _____ is reduced using 9. _____ to form 10. _____. One molecule of 11. _____ leaves the cycle as a final product, while the other five molecules continue through the Calvin cycle. The five molecules go through a series of reactions to become 12. _____, which is then used to fix more 13. _____. The 14. _____ molecule released from the Calvin cycle is used to form 16. _____. This can be stored as 17. _____ molecules.

NADPH	RuBP	ATP	Glucose
	CO2	PGA	G3P
phosphorylates	3-phosphoglycerate	1, 3-biphosphoglycerate	

What I Can Do

Directions: Make a presentation to show the steps involved in Calvin Cycle. Choose one type of presentation from the options below:

1. Compose a song/jingle about Calvin Cycle.
2. Construct a diagram showing the steps involved in Calvin Cycle.
3. Simulate Calvin Cycle. (Digital Simulation)
4. Make a poem/short story about Calvin Cycle.
5. Present Calvin Cycle through images and symbol.

Assessment

Directions: Write the letter of the correct answer.

___1. Which of the following molecule combines with CO_2 at the beginning of the Calvin cycle?
A. RuBP B. PGA C. 3-G3P D. NADPH

___2. Which of the following molecule is produced for every three molecules of CO that enter the Calvin cycle?
A. RuBP B. ATP C. 3-PGA D. NADPH

___3. Which of the following are end-products of Calvin cycle?
A. lipids. C. fatty acids
B. amino acids. D. carbohydrates

___4. Which is the correct order for the stages of the Calvin cycle?
A. Carboxylation → Regeneration → Reduction
B. Regeneration → Carboxylation → Reduction
C. Reduction → Carboxylation → Regeneration
D. Carboxylation → Reduction → Regeneration

___5. Which of the following best describes Calvin Cycle?
A. occur only in the dark phase
B. produce the reduced coenzyme NADH
C. main source of ATP in photosynthesis
D. convert carbon dioxide to carbohydrates

What I Need to Know

Cellular respiration is a common idea that animals as well as humans need to use oxygen in their breathing. This is an essential biological mechanism for living organisms. There are two types of respiration which one requires oxygen, and the other needs no oxygen.

At the end of the lesson the learners are expected to:
Differentiate aerobic from anaerobic respiration (STEM-BIO 12-IIa-j-6)

What I Know

Directions: Choose the letter of the correct answer.

_____ 1. What is/are the product/s of aerobic respiration?
A. lactic acid C. glucose and oxygen
B. carbon monoxide D. carbon dioxide and water

_____ 2. What is/are the product/s of anaerobic respiration?
A. lactic acid C. glucose and oxygen
B. carbon monoxide D. carbon dioxide and water

_____ 3. What type of respiration requires oxygen?
A. aerobic C. anaerobic
B. exhalation D. inhalation

_____ 4. What type of respiration does not require oxygen?
A. aerobic C. anaerobic
B. exhalation D. inhalation

_____ 5. Which types of respiration are most efficient?
A. aerobic C. anaerobic
B. exhalation D. inhalation

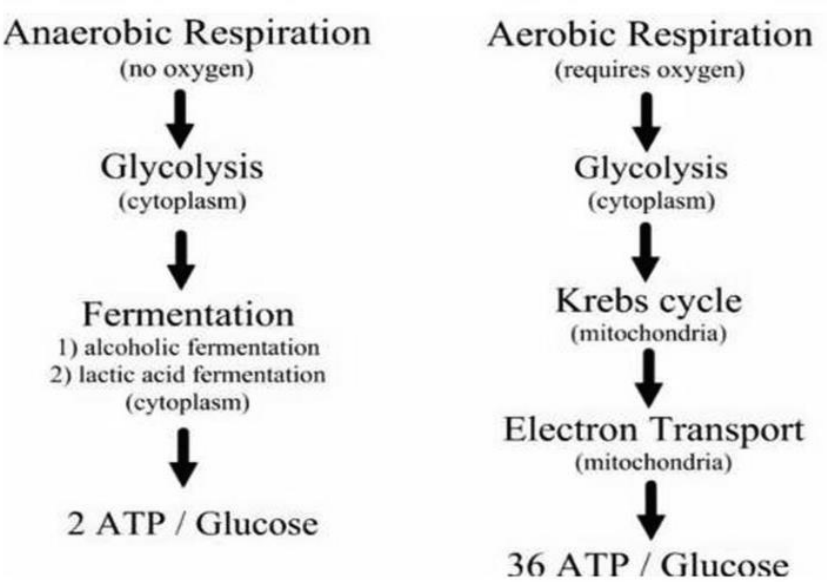
Lesson 4	*Differentiate aerobic from anaerobic respiration*

What's New

Activity

What's In, What's Out?

Directions: Analyze the flow chart and fill in the table with the needed information.



Criteria	Aerobic Respiration	Anaerobic Respiration
Oxygen used		
Definition		
Process involved		
Products		
Parts of the cells where it takes place		

What Is It

There are two types of cellular respiration: aerobic respiration takes place in the presence of oxygen and in the absence of oxygen, anaerobic respiration happens. Both processes start with glycolysis. **Glycolysis** is an anaerobic process where oxygen is not needed, splitting of glucose into three molecules of carbon. A minimal amount of ATP is generated by this process. The **Krebs cycle and electron transport** require oxygen to continue, and these processes generate much more ATP than glycolysis alone in the presence of oxygen. In the cytoplasm of the cell glycolysis occurs while the other aerobic respiration processes take place in the mitochondria.

The presence of oxygen decides which products are produced. Carbon dioxide, water, and ATP are products of aerobic respiration from food such as

carbohydrates, lipids, and protein. Lactic acid, ethanol and ATP are products of anaerobic respiration.

What's More

Directions: Fill in the graphic organizer.

AEROBIC RESPIRATION	ANAEROBIC RESPIRATION
How alike?	
AEROBIC RESPIRATION	**ANAEROBIC RESPIRATION**
How different?	

What I Can Do

List five (5) benefits of anaerobic and aerobic exercises to human system.

_____

Assessment

Directions: Fill in the blanks to complete the statement.

1. Respiration makes _____ available to all _____.
2. Aerobic respiration uses _____.
3. _____ respiration occurs in the cytoplasm of cell.
4. Carbon dioxide and water are the products _____ respiration
5. _____ respiration occurs in the mitochondria of the cell.

What I Need to Know

Cellular respiration happens over three major pathways such as **glycolysis**, **citric acid cycle**, and **oxidative phosphorylation**. It occurs in the cytoplasm and mitochondria of the cell.

At the end of the lesson, learners are expected to:

Explain the major features and sequence the chemical events of cellular respiration (STEM-BIO 12-IIa-j-7)

What I Know

Directions: Write TRUE if the statement is correct and FALSE if it is not.

_____1. All forms of life need energy.
_____2. ATP is the "energy currency" of the cell.
_____3. Plant cell has mitochondria.
_____4. Cellular Respiration occurs in chloroplast.
_____5. Most living things use glucose to make ATP from oxygen.

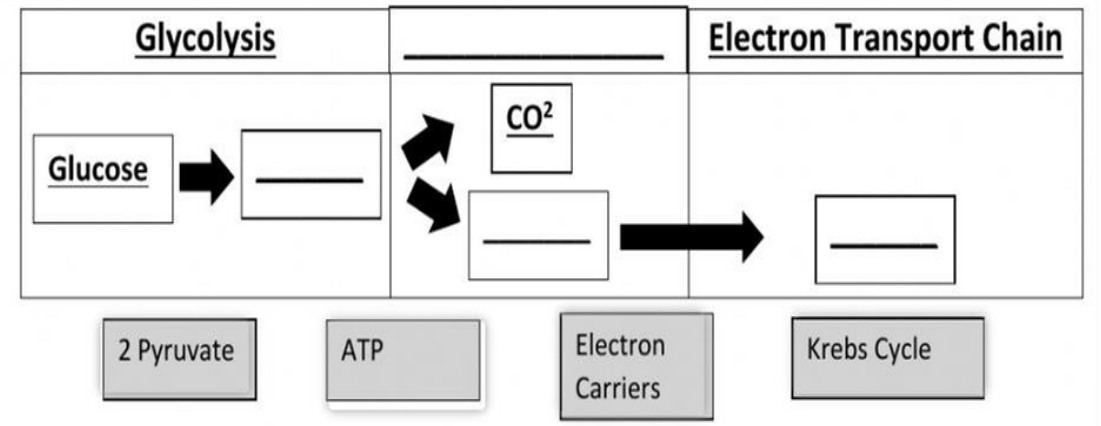
Lesson 5 — *CELLULAR RESPIRATION*

What's New

Activity

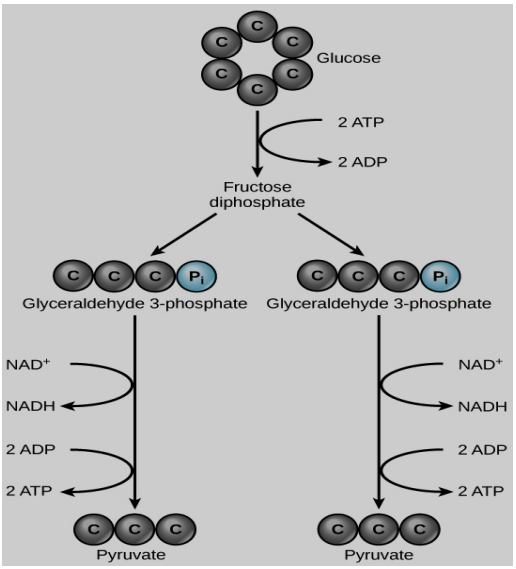
Break Out

Directions: Pick the right term in the box below to complete the process.

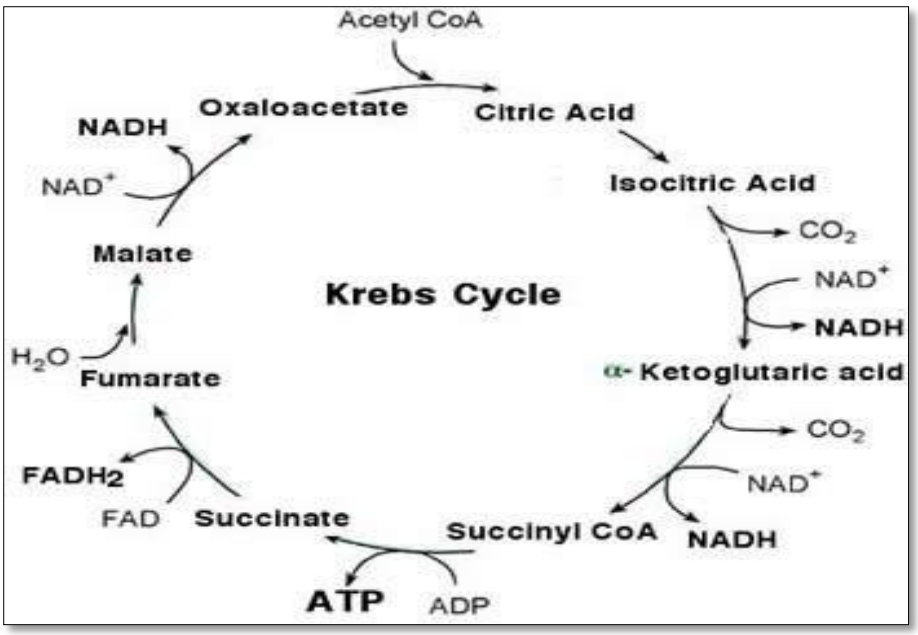


What is It

Glycolysis occurs in cytoplasm, a process by which glucose molecules split into two pieces called pyruvate acid, a three-carbon organic molecule. It requires 10 enzymes to happen which catalyze each of the ten steps, as well as an investment of two ATP molecules in the preparatory phase or energy consumption phase to get four ATP back over several steps in the payoff phase or energy producing phase. During energy consumption phase, one glucose (simple sugar) molecule is broken down into two glyceraldehyde 3-phosphate or simply PGAL. The net ATP gain



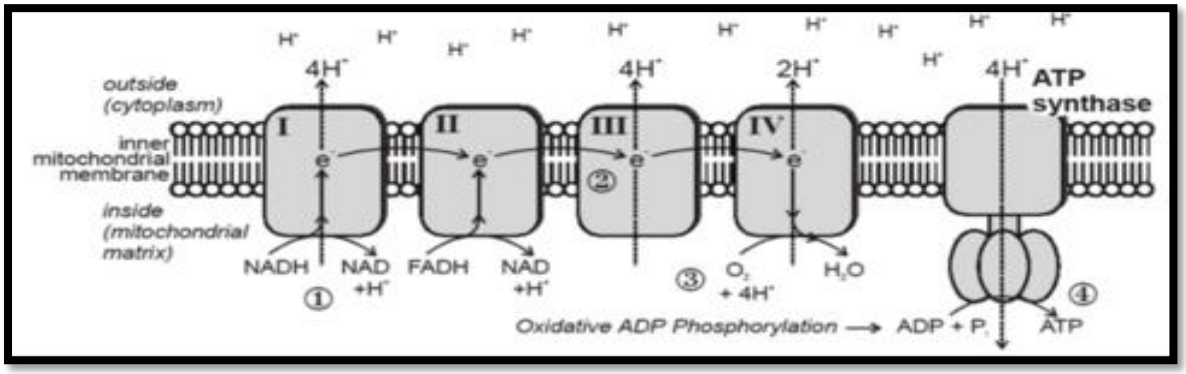
from this is only two ATP per glucose molecule, glycolysis also produces two pairs of high-energy electrons, which are passed on to an electron carrier called nicotinamide adenine dinucleotide (NAD*). It converts NAD+ into NADH until the phase completes the conversion of two PGAL into pyruvate acid.



Krebs Cycle or Citric Acid Cycle is the second step in cellular respiration which needs oxygen to proceed to a series of chemical reactions wherein the bonds of the pyruvic acids are broken apart. It takes place in the inner matrix of mitochondria. The pyruvate that is being produced during glycolysis is taken in inner membrane to start the process, which is converted into acetyl-CoA and releases carbon dioxide. Acetyl- CoA combines with another compound forming citric acid.

Citric Acid is continuously broken down in the Krebs Cycle. The breaking apart of citric acid releases carbon dioxide (CO_2) as a by-product. The series of chemical reactions lead to formation of products of one reaction and become the essential compounds for the next reaction. During the Krebs Cycle, high-energy bonds are pulled apart and high energy electron carriers, namely NAD+ and flavin adenine dinucleotide (FAD). Each of this electron carriers can accept a pair of high-energy electrons. These electron carriers hold the electrons for the short period of time before passing them on to another compound. They remove pairs of high-energy electrons during one turn of the Krebs Cycle.

As the Krebs Cycle started, pyruvate molecules are being produced in glycolysis, two pyruvate molecules that double the amount of product produced during the Krebs Cycle to have an accurate estimate of how much energy can produce in one glucose molecules during this stage. The grand total for this reaction is two ATP, six NADH molecules and two FADH2 molecules. Electrons from glycolysis and Krebs Cycle will be used on final stage of cellular respiration called **Electron Transport Chain (ETC).**

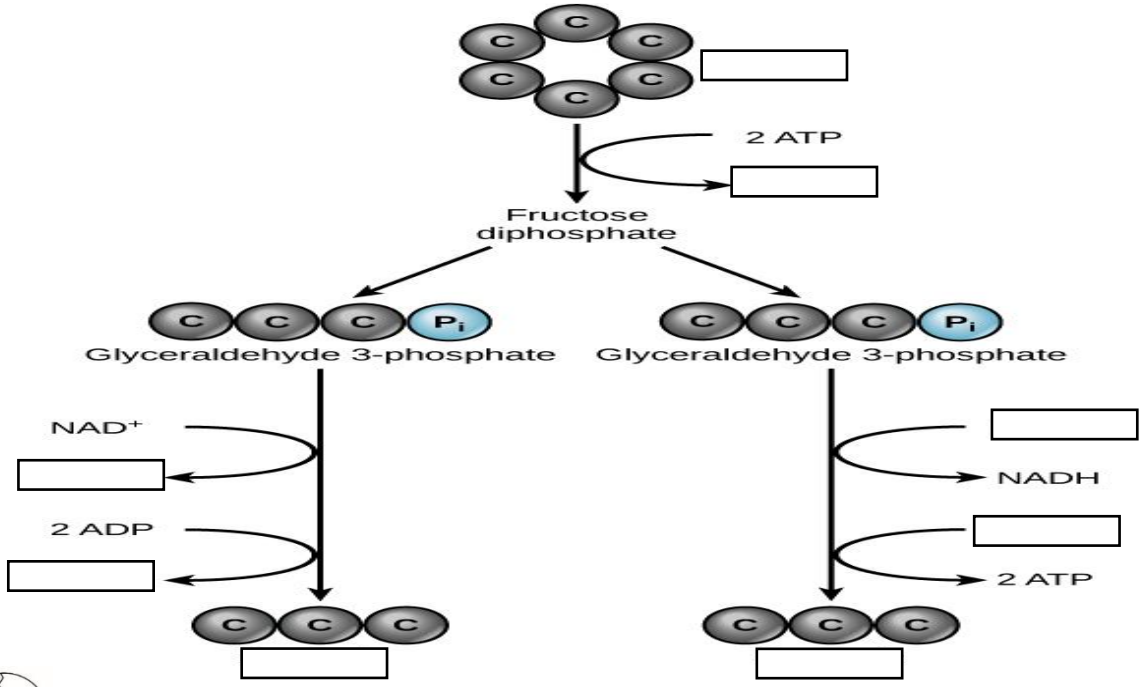


The **Electron Transport Chain (ETC)** is the next step wherein the electron carriers (NAD+ and FAD+) transport the high-energy electrons for conversion of ADP to ATP. The high-energy electrons transfer through four sets of enzyme complexes which are named into I, II, III and IV of the inner mitochondria membrane. This movement of electrons generates energy that is being used in pumping hydrogen ions into intermembrane space from the matrix of the mitochondria. This continues pumping of hydrogen ions causes the increase of concentration in intermembrane space of mitochondria. The ATP synthase converts the energy of protons (H+) moving down their concentration gradient into the synthesis of ATP. There are 3 to 4 protons moving through this machine is enough to convert a molecule of ADP and Pi (inorganic phosphate) into a molecule of ATP. And the oxygen is the final electron acceptor that takes a pair of hydrogen ions result of the formation of water molecules.

What's More

Activity **Glycomplete**

DIRECTIONS: Supply the right term in each box to complete the diagram.



What I Can Do

DIRECTIONS: Write the appropriate terms to complete the process.

1. The high-energy electrons by NAD+ and FAD+ transporter through the process called _____.
2. The energy from the electrons travel along the four complexes is used to pump hydrogen ions across _____.
3. The build-up hydrogen ions making the intermembrane space positively charge while the _____becomes negative charge.
4. The _____convert the energy of protons (H+) moving down their concentration gradient into the synthesis of ATP _____.
5. The energy from the moving ions used to combine ADP and _____ forming a high –energy ATP.

Assessment

DIRECTIONS: Write the letter of the correct answer.

_____1. What is the correct sequence of the steps of cellular respiration?
A. Krebs Cycle, Glycolysis, Electron Transport Chain
B. Electron Transport Chain, Glucose, Krebs Cycle
C. Krebs Cycle, Electron Transport Chain, Glycolysis
D. Glycolysis, Krebs Cycle, Electron Transport Chain

_____2. Where does Krebs cycle take place?
A.cristae C. thylakoid membrane
B.matrix D. intermembrane space

_____3. Where does the Electron Transport Chain (ETC) take place?
A. matrix C. intermembrane space
B. thylakoid membrane D. cristae of the inner membrane

_____4. Where does glycolysis take place?
A. matrix C. intermembrane space
B. cytoplasm D. cristae of the inner membrane

_____5. How many pyruvates acid produced in Glycolysis?
A. one B. two C. four D. three

What I Need to Know

In general, a cell contains only enough ATP to sustain a 30-second to few minutes of normal activity. Because of a high potential energy, ATP is unstable and is not stored. Most of the ATP produce used the chemical energy from glucose.

At the end of the lesson learners are expected to:
Distinguish major features of glycolysis, Krebs cycle, electron transport system, and chemiosmosis (STEM-BIO 12-IIa-j-8)

What I Know

Directions: Write the letter of the correct answer.

_____1. Which cellular respiration occurs in the presence of oxygen?
A. glycolysis C. aerobic respiration
B. oxygen respiration D. anaerobic respiration

_____ 2. Which of the following is an example of anaerobic respiration?
A. glycolysis C. Calvin – Benson cycle
B. citric acid cycle D. electron transport chain

_____ 3. Which process converts glucose into two molecules of pyruvate?
A. Krebs cycle C. Pyruvate processing
B. Light Reaction D. Calvin – Benson Cycle

_____ 4. During glycolysis, phosphoenolpyruvate is converted into pyruvate. Which enzyme does the conversion?
A. enolase C. phosphoglycerate mutase
B. pyruvate kinase D. phosphoenolpyruvate kinase

_____ 5. Which molecule is at the ATP investment stage of glycolysis?
A. phosphoenolpyruvate C. dihydroxyacetone phosphate
B. 6 glucose – 6- phosphate D. glyceraldehyde – 3 – phosphate

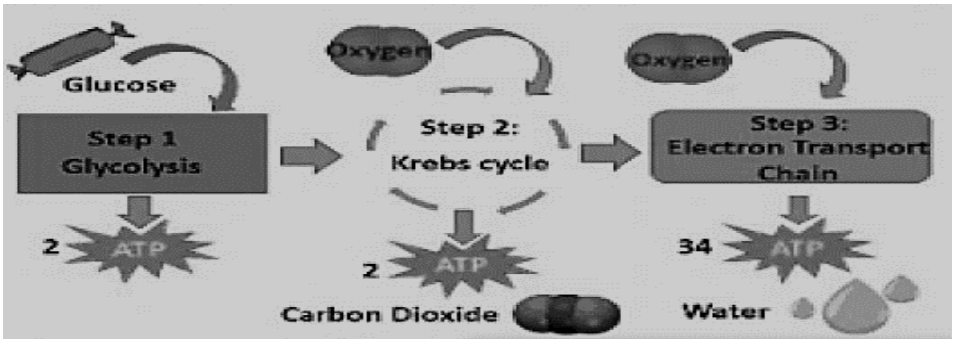
Lesson 6	*Important Features of Stages of Cellular Respiration*

What's New

Activity

Take My Breath Away

Directions: Use the illustration to answer the guided questions.



Guided Questions:
1. Which step uses the glucose? oxygen? _____
2. Which step does the waste product carbon dioxide come from? _____
3. Based on the image, which step is the most important when it comes to supplying ATP?

What Is It

Photosynthetic organisms can produce glucose from the products of photosynthesis, where the energy in sunlight is used to reduce carbon dioxide (CO_2). These organisms will either use the glucose to make ATP or store it in other energy-rich molecules like starch. When photosynthetic organisms are eaten or decompose, their glucose molecules are obtained by animals, fungi, and many bacteria and archaea. Storage carbohydrates, such as starch and glycogen, act like savings accounts for chemical energy. ATP, in contrast, is like cash. To withdraw chemical energy from the accounts to get cash, storage carbohydrates are first hydrolyzed into their glucose monomers. The glucose is then used to produce ATP through one of two general processes: cellular respiration or fermentation. The primary difference between these two processes lies in the degree to which glucose is oxidized. When glucose undergoes the uncontrolled oxidation reaction called burning, some of the potential energy stored in its chemical bonds is converted to kinetic energy in the form of heat and light. **Fermentation** is another process that does not fully oxidizes glucose. small, reduced organic molecules are produced as waste. The complete oxidation of glucose via cellular respiration as a set of four interconnected processes that together convert the chemical energy in glucose to chemical energy in ATP. Each of the four processes consists of a distinctive starting molecule, a series of chemical reactions, and a characteristic set of products.

1. Glycolysis

During glycolysis, one six-carbon molecule of glucose is broken into two molecules of the three-carbon.

compound pyruvate. During this process, ATP is produced from ADP and Pi, and nicotinamide adenine dinucleotide (NAD+) is reduced to form NADH.

2. Pyruvate processing

Each pyruvate is processed to release one molecule of CO2, and the remaining two carbons are used to form the compound acetyl CoA. The oxidation of pyruvate results in more NAD+ being reduced to NADH.

3. Citric acid cycle

Each acetyl CoA is oxidized to two molecules of CO2. During this sequence of reactions, more ATP and NADH are produced, and flavin adenine dinucleotide (FAD) is reduced to form FADH2.

4. Electron transport and oxidative phosphorylation

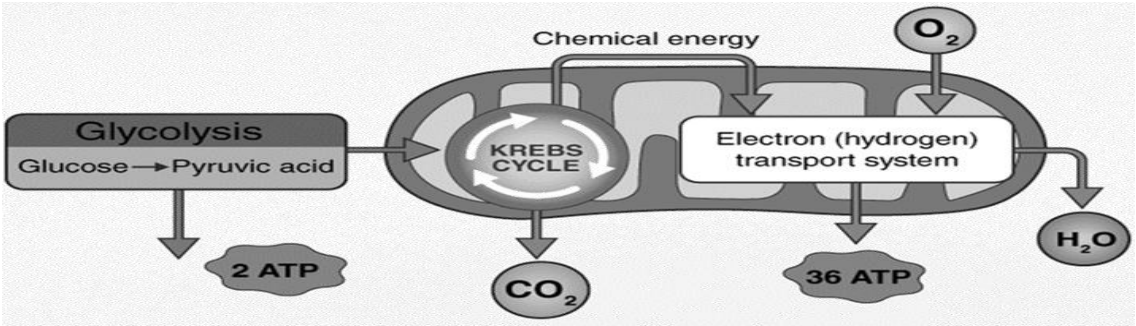
Electrons from NADH and FADH2 move through a series of proteins that together are called an electron transport chain (ETC). The energy released in this chain of redox reactions is used to create a proton gradient across a membrane; the ensuing flow of protons back across the membrane is used to make ATP. Because this mode of ATP production links the phosphorylation of ADP with the oxidation of NADH and FADH2, it is called oxidative phosphorylation.

What's More

Activity

2-2-34

Directions: Study the illustration and answer the guided questions.



Guided questions:

1. How many ATP molecules are added to get glycolysis started? _____
2. Since glycolysis produces 4 ATP molecules, this results in a NET Gain of _____ ATPs.
3. What is the total ATP produced? _____

What I Can Do

Directions. Answer the following questions.

1. We inhale oxygen when we breathe and exhale carbon dioxide. What is the oxygen used for and where does the carbon dioxide come from?

_____

2. When muscle cells run out of oxygen, what happens to the potential for energy extraction from sugars and what pathways do the cell use?

_____

Assessment

Directions. Write the letter of the correct answer.

_____ 1. Krebs cycle transforms Acetyl CoA into numerous acids. Which of the following happens when succinic acid transforms into fumaric Acid?
A. It will fuse to pyruvic acid to produce lactic acid
B. It will split into another RuDP and a sugar molecule
C. Two FAD will absorb 2 H to produce 2 FADH2
D. It will release 2 CO2 & will produce 2 NADH + 2H+

_____ 2. Where is the electron transport chains of cellular respiration located?
A. in the matrix of the mitochondrion
B. in the inner membrane of the mitochondrion
C. in the outer membrane of the mitochondrion
D. in the intermembrane space of the mitochondrion

_____ 3. Which of the following process explains chemiosmosis?
A. the movement of electrons across the cell membrane
B. the movement of glucose through the cell membrane
C. the movement of hydrogen atoms across a mitochondrial membrane
D. the movement of hydrogen ions across a mitochondrial membrane

_____ 4. Which process will follow if oxygen is not present during glycolysis?
A. respiration
B. fermentation
C. photosynthesis
D. electron transport chain

_____ 5. Which pathway produces the most ATP per glucose molecule?
A. glycolysis
B. Krebs cycle
C. fermentation
D. electron transport chain

What I Need to Know

Adenosine triphospate (ATP) is the energy currency for cellular processes. It provides the energy for both endergonic and exergonic reactions which require small input of activation energy. On the other hand, oxygen is essential to have efficient cellular respiration; most organisms need oxygen for a single purpose: to release energy from food for use by cells.

At the end of the lesson the learners are expected to:

1. Describe reactions that produce and consume ATP (STEM-BIO 12-IIa-j-9)

2. Describe the role of oxygen in respiration and describe pathways of electron flow in the absence of oxygen (STEM-BIO 12-IIa-j-10)

What I Know

Directions: Write the letter of the correct answer.

_____1. Which of the following reactions describes as the energy from sunlight drives the synthesis of ATP and NADPH, coupled to the formation of O_2 from H_2O?
A. dark reaction
B. light reaction
C. endergonic reaction
D. exergonic reaction

_____2. Which of the following is NOT included in the major reactions' pathway in cellular respiration?

A. Calvin cycle
B. electron transport chain
C. glycolysis
D. Krebs cycle

_____3. What is the formation of ATP by the direct transfer of a PO_3 group to ADP?

A. chemiosmosis
B. phosphorylation
C. oxidative phosphorylation
D. substrate-level phosphorylation

_____4. What is the role of oxygen in cellular respiration?

A. absorbs energy
B. enhance oxidation
C. carrier of electrons
D. final acceptor in ETC

_____ 5. What reaction uses the energy from photons to create ATP and NADPH, which are both forms of chemical energy used in the dark reaction?

A. dark Reaction
B. light Reaction
C. endogenic Reaction
D. exergonic Reaction

Lesson 7	***Role of ATP and Oxygen in Respiration***

What's New

Activity

You and Me

Directions: Compare and contrast ATP and glucose to a transportation tank.

ATP and glucose are both molecules that organisms use for energy. They are like the tank of a delivery truck that brings gas to a gas station and the gas tank that holds the fuel for a car. Which from ATP and glucose molecules has more energy storage? Explain your answer. _____

What Is It

During photosynthesis energy from sunlight is harvested and used to drive the synthesis of glucose from CO_2 and H_2O in the cytoplasm. By converting the energy of sunlight to a usable form of potential chemical energy, photosynthesis is the ultimate source of metabolic energy of all biological systems. This process is composed of two distinct stages. In light reaction, energy from the sunlight drives the synthesis of ATP and NADPH, coupled to the formation of O_2 from H_2O while in dark reaction the ATP and NADPH produced by the other reactions drive glucose synthesis. On the other hand, during the cellular respiration actually "burns" glucose for energy. However, it doesn't produce light of intense heat as some other types of burning do. This is because it releases the energy in glucose slowly, in many small steps. It uses the energy that released to form molecule of ATP.

Carbon dioxide is a fully oxidized molecule. Basically, it does not have a lot of chemical energy. The Calvin cycle turns carbon dioxide into the much more reduced molecule, glucose, which has much more energy.

Significance of oxygen to cellular respiration:

- Oxidize food to release energy during respiration

- Final acceptor in the Electron Transport Chain (ETC). It is important because it ensures that there aren't too many free-floating electrons and protons, which are damaging to cells
- Without ATP from mitochondria, cells would only have access of ATP from fermentation (and from glycolysis), which is sadly inadequate for most eukaryotic cells.
- Without the heat "usable" energy "leaking" from continuous "slow hydrogen combustion", body temperature would quickly fall, with easily predictable long-term results.

What's More

ACTIVITY **OXYGEN BANKER**

Directions: Using the word bank supply the needed information to complete the paragraph.

Word Bank:

NAD^+, final electron acceptor, oxygen, electrons, NADH

When no _____ is present, **the electron transport chain** can't run because there is no oxygen to act as the _____. This means that the ETC will not be accepting _____ from _____ as its source of power, so _____$^+$ will not be regenerated.

What I Can Do

How will oxygen affect biological processes when it is not around?

_____

Assessment

Directions: Write PS if the statement is related to photosynthesis while CR if associated to cellular respiration.

_____1. Carbon dioxide and water combine in the presence of sunlight to produce glucose and oxygen.
_____2. It produces 38 ATP molecules per glucose, but actual yield is only about 30-32.
_____3. Carbon dioxide is absorbed, and oxygen is released.
_____4. Glucose is broken down into water and carbon dioxide and energy.
_____5. The production of organic carbon (glucose and starch) from inorganic carbon (carbon dioxide) with the use of ATP and NADPH produced in the light dependent reaction.

What I Need to Know

Two fundamental cellular processes are cellular respiration and photosynthesis. Cellular respiration is the process by which cells release stored energy from sugars. Photosynthesis is the process in which producer cells use carbon dioxide, water, and nutrients to produce glucose and oxygen.

Respiration is essential for survival of living organisms. It releases energy from the food. The oxygen we inhale is used to breakdown glucose into carbon dioxide and water. Energy is released in the process. ATP is the main source of energy in our body. Our body entirely depends upon the energy gained from the food that we eat.

At the end of the lesson the learners are expected to:

1. **Compute the number of ATPs needed or gained in photosynthesis and respiration (STEM-BIO12-IIa-j-11); and:**
2. **Explain the advantages and disadvantages of fermentation and aerobic respiration. (STEM-BIO12-IIa-j-12)**

What I Know

Directions: Write the letter of the correct answer.

_____1. How many molecules of ATP are needed for glycolysis to take place?
A. 4 B. 3 C. 2 D. 1

_____2. At the end of glycolysis, there is a net gain of how many ATP molecules?
A. 4 B. 3 C. 2 D. 1

_____3. What are the products of an aerobic respiration?
A. glucose and oxygen C. glucose and water
B. carbon dioxide and water D. carbon dioxide and oxygen

_____4. What are the products of an anaerobic respiration?
A. lactic acid C. carbon dioxide and water
B. glucose and oxygen D. carbon dioxide and oxygen

_____5. Which organelle does an aerobic respiration take place?
A. nucleus B. cell membrane C. mitochondia D. cytoplasm

Lesson 8 *Advantages and Disadvantages of Fermentation and Aerobic Respiration*

What's New

ACTIVITY A: ATP REQUIREMENT AND GAINED IN CELLULAR RESPIRATION

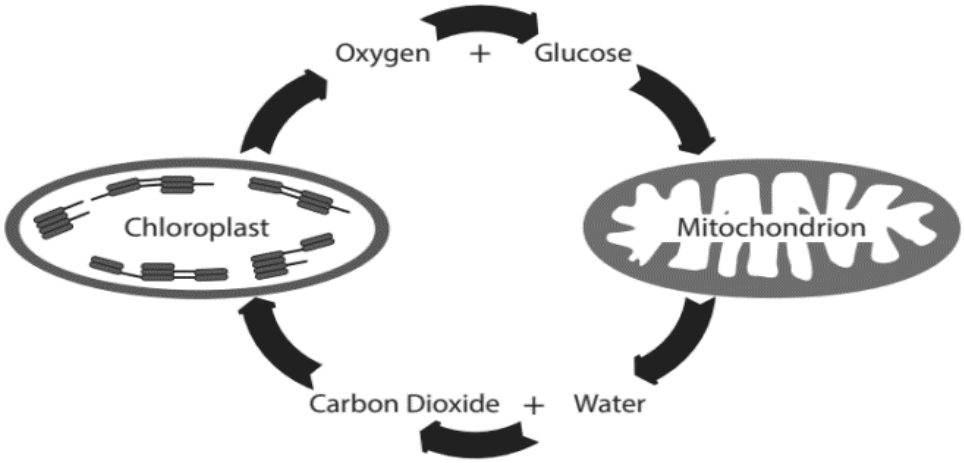
Directions: Fill in the number of ATP's required and gained through the process of cellular respiration

Stages in Cellular Respiration	Location	Reactants	Number of ATP Used	Major Products	Secondary Products	Number of ATP's Produced
Glycolysis	Cytoplasm	1 glucose molecule		2 Pyruvate molecules	2 NADH	

Pyruvate Oxidation (Link Reaction)	mitochondrial matrix	2 pyruvate molecules		2 Acetyl-CoA	2 NADH CO_2	
Kreb Cycle (Citric Acid Cycle or Tricarboxylic Acid Cycle)	mitochondrial matrix	2 Acetyl-CoA		6 NADH 2 $FADH_2$ (high energy electron carriers)	CO_2	
Oxidative phosphorylation (electron transport chain and chemiosmosis)	folded inner membrane of the mitochondria (cristae)	6 NADH 2 FADH2 (high energy electron carriers)		ATP molecules	H_2O	
TOTAL						

ACTIVITY B: Photosynthesis and Cellular Respiration

Directions: Write the requirements needed and products of both processes in the designated box.



CELLULAR PROCESSES	REQUIREMENTS (Reactants)	PRODUCTS
Photosynthesis		
Cellular Respiration		

What Is It

A. Connecting Cellular Respiration and Photosynthesis

Trees release oxygen as a byproduct of photosynthesis, and you need oxygen to breathe. So, your cells can perform cellular respiration and make ATP. Photosynthesis and cellular respiration are connected through an important relationship. This relationship enables life to survive as we know it. The products of one process are the reactants of the other. Notice that the equation for cellular respiration is the direct opposite of photosynthesis.

Cellular Respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$
Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Photosynthesis makes the glucose that is used in cellular respiration to make ATP. The glucose is then turned back into carbon dioxide, which is used in photosynthesis. While water is broken down to form oxygen during photosynthesis, in cellular respiration oxygen is combined with hydrogen to form water. While photosynthesis requires carbon dioxide and releases oxygen, cellular respiration requires oxygen and releases carbon dioxide. It is the released oxygen that is used by us and most other organisms for cellular respiration. We breathe in that oxygen, which is carried through our blood to all our cells. In our cells, oxygen allows cellular respiration to proceed. Cellular respiration works best in the presence of oxygen. Without oxygen, much less ATP would be produced.

Cellular respiration and photosynthesis are important parts of the carbon cycle. The carbon cycle is the pathways through which carbon is recycled in the biosphere. While cellular respiration releases carbon dioxide into the environment, photosynthesis pulls carbon dioxide out of the atmosphere. The exchange of carbon dioxide and oxygen during photosynthesis and cellular respiration worldwide helps to keep atmospheric oxygen and carbon dioxide at stable levels.

B. Advantages and Disadvantages of Fermentation and Aerobic Respiration

Aerobic and anaerobic respiration each have advantages under specific conditions. Aerobic respiration produces far more ATP, but risks exposure to oxygen toxicity. Anaerobic respiration is less energy-efficient but allows survival in habitats which lack oxygen.

The reactions of glycolysis have no specific requirement for oxygen. Oxidation reactions do occur. Under anaerobic conditions, pyruvate may be reduced to a variety of different compounds. Alcoholic fermentation is a common pathway in microorganisms and is of industrial importance.

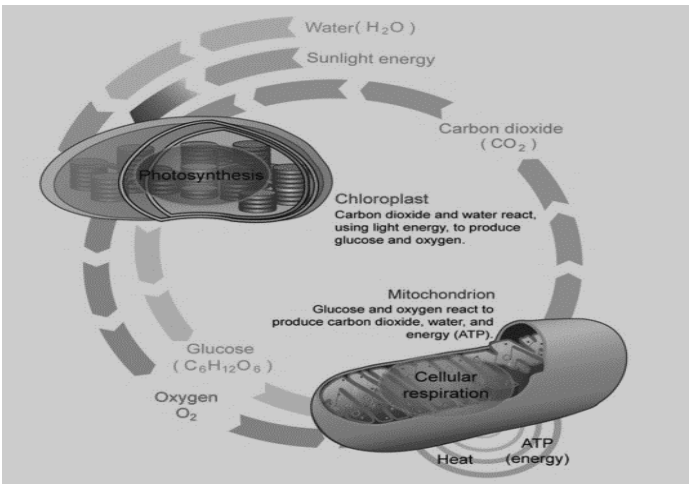
What's More

A. Photosynthesis and Cellular Respiration: A Direct Opposite Cycle

Directions: Draw a diagram showing the cycle of photosynthesis and cellular respiration as a direct opposite cycle, label it and write a short discussion. (Use a separate sheet of paper).

B. Anaerobic and Aerobic Respiration: A Comparison

Directions: Fill in the table with advantages and disadvantages of both aerobic and anaerobic respiration. (Use separate sheet of paper if needed)



TYPES OF RESPIRATION	ADVANTAGES	DISADVANTAGES
AEROBIC		
ANAEROBIC		

What I Can Do

Directions: Analyze the given situation and answer the question.

A compound made from lactic acid is used as a food preservative, but the applications of lactic acid extend far beyond food production. Lactic acid is important as a starting material for making drugs in the pharmaceutical industry. Additionally, it is involved in the manufacturing of lacquers and inks; is used as a humectant, or moisturizer, in some cosmetics. As a consumer, how do we ensure that these are not harmful to us? _____
_____
_____.

Assessment

Directions: Write the letter of the correct answer.

_____1. What is the fuel used by cells in cellular respiration to get energy?
A. sugar B. glucose C. carbon dioxide D. lactic acid

_____2. Which kind of cellular respiration which occurs in the presence of oxygen?
A. nitrogenous respiration C. anaerobic respiration
B. aerobic respiration D. acidic respiration

_____3. Which of the following is the chemical formula for cellular respiration?
A. glucose + oxygen $\longrightarrow$ carbon dioxide + water
B. carbon dioxide + oxygen $\longrightarrow$ oxygen + water
C. carbon dioxide + water $\longrightarrow$ oxygen + water + ATP
D. glucose + oxygen $\longrightarrow$ carbon dioxide + water + ATP

_____4. If yeast was used to make a bread dough rise, what kind of fermentation did the process undergo?
A. acetic fermentation C. glucose fermentation
B. alcohol fermentation D. lactic acid fermentation

_____5. Which type of fermentation does bacteria and yeast were used to make wine and beer?
A. acetic fermentation C. glucose fermentation
B. alcohol fermentation D. lactic acid fermentation

Haiku Poetry Rubric

Category	5 points	3 points	1 point	0 points
Haiku poetry form	The poem has 3 lines, with the proper number of syllables in each line:5,7,5	The poem has 3 lines, 2 of which have the correct syllable count.	The poem has 3 lines, but only 1 line has the correct number of syllables.	The poem has 3 lines, but none has the correct syllable count.
Structure: line relationship	Lines relate to each other;	Second line does not relate to first; third line relates to second.	Second line does not relate to first, and third line does not relate to second.	Lines do not appear to be about the same subject.
Subject	Poem presents a laboratory safety rule, situation, or procedure	Poem partially presents a laboratory safety rule, situation, or procedure	Poem marginally presents a laboratory safety rule, situation, or procedure	Poem fails to present a laboratory safety rule, situation, or procedure
Neatness and spelling	Neatly written and centered. No spelling errors.	Neatly written but not centered. 1 spelling error	Not neatly written. More than 2 spelling errors	Smudged, smeared; messy. 3 or more spelling errors

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